Practice Set 10 Paper 1 – Basic Sciences in Urology

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Practice paper compiled from the question bank.

Q1. Management of chyluria

Type: LAQ (25 marks) (also answered in 2024-Winter Paper 3, Q2 and 2018-Summer Paper 1, Q1)

Definition and mechanism

- **Lymphatic anatomy:** intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction. Renal lumbar lymphatics join the cisterna; valves normally prevent backflow.
- **Chyluria** is the passage of **chyle** (intestinal lymph rich in chylomicrons and protein) in urine, which looks milky.
- It needs a **lymphourinary fistula**, usually a **pyelolymphatic fistula** at a calyceal fornix.
- **Pathogenesis (filarial):** adult *Wuchereria bancrofti* worms and the host reaction cause **lymphangitis and fibrosis** between the intestinal trunks, cisterna chyli and thoracic duct. Lymphatics below the block dilate and their valves fail. Chyle refluxes into renal hilar lymphatics, which rupture into a calyx.
- It appears **earlier than elephantiasis**, is usually **intermittent** and may remit spontaneously. So every treatment is judged against natural remission.

Causes

Parasitic (commonest; India, China)	Non-parasitic
W. bancrofti (lymphatic filariasis, Culex mosquito)	Congenital lymphangiectasia, renal lymphangioma
	Trauma or surgery (partial nephrectomy, RPLND [retroperitoneal lymph node dissection])
	Retroperitoneal tumour, fibrosis, tuberculosis, pregnancy

Clinical features and severity

- Milky urine, worse after fatty meals; **chylous clots** (colic, retention); **haematochyluria**.
- Loss of fat and protein: weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.

Grade	Features
Mild	Intermittent; no systemic effect
Moderate	Persistent; clots
Severe	Haematochyluria, clot retention, weight loss, hypoalbuminaemia

Evaluation (before treatment)

1. Confirm chyle

- **Ether (chloroform) test:** urine shaken with ether clears, because the fat dissolves.
- Urine **triglycerides** and protein raised; Sudan III stains fat globules.
- Exclude **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial status

- Night blood smear (often negative late); **circulating filarial antigen** card test; eosinophilia; PCR (polymerase chain reaction).

3. Lateralise and localise the fistula

- **Cystoscopy after a fatty meal:** milky efflux from one ureteric orifice (sometimes both).

- **Retrograde pyelography:** pyelolymphatic backflow.
- **Lymphoscintigraphy** (non-invasive); MR (magnetic resonance) lymphangiography; pedal lymphangiography is historical.

4. Baseline: albumin, haemoglobin, renal function; urine culture before instillation.

Management algorithm

```
Confirm chyle -> grade severity
   Mild             -> diet + antifilarial drugs; observe
   Moderate/persist -> add endoscopic sclerotherapy (repeatable)
   Severe or failed -> renal hilar lymphatic disconnection
```

1. Conservative (all patients)

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil** and plenty of fluids.
- **Why MCT:** medium-chain fats are absorbed directly into portal blood, not into lacteals, so less chyle is formed.
- **DEC (diethylcarbamazine)** 2 mg/kg orally three times daily, traditionally 12 days; a 1-day course is as effective. It kills microfilariae but has only modest effect on adults.
- **Doxycycline** 200 mg daily for 4–8 weeks kills *Wolbachia* (endosymbiont) and so sterilises or shortens the life of adult worms. Albendazole and ivermectin are alternatives.
- Correct anaemia and protein deficit. Most mild cases settle on this, and many remit spontaneously.

2. Endoscopic sclerotherapy (renal pelvic instillation sclerotherapy)

- Ureteric catheter on the leaking side, head-down position. Instil **silver nitrate 0.5–1%** or **povidone-iodine**; repeat over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- **Contraindications:** active UTI (urinary tract infection), bilateral simultaneous instillation (risk of anuria).
- Good initial response; recurrence is common, but it is repeatable and avoids surgery.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death with silver nitrate.

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:**
  1. Full renal mobilisation (nephrolysis).
  2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every hilar lymphatic.
  3. Strip the upper ureter; remove perinephric fat.
  4. Nephropexy; optional **omental wrap** around the hilum as a barrier.
  5. **On-table cystoscopy** confirms clear efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open in meta-analysis: shorter operation and stay, less blood loss, faster bowel recovery, fewer complications.
- **Complications:** lymphocele, renal vein injury, recurrence.
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy now obsolete.

4. Follow-up and prevention

- Urine triglycerides and weight; recurrence needs repeat sclerotherapy or surgery.
- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: retroperitoneal lymphatics with obstructed thoracic duct, dilated hilar lymphatics and pyelolymphatic fistula at a calyceal fornix.

Key points to remember:

- Pyelolymphatic fistula from *W. bancrofti* lymphatic obstruction; often intermittent.
- Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
- Diet with MCT plus DEC → silver nitrate sclerotherapy → hilar lymphatic disconnection.
- Retroperitoneoscopic disconnection beats open.

Q2. Physiological and hemodynamic effects of laparoscopy. Write in detail about lap donor nephrectomy

Type: LAQ (25 marks) (also answered in Practice Set 3 Paper 1, Q1 and Practice Set 2 Paper 4, Q3a)

Part A: Physiological and haemodynamic effects of laparoscopy

- **Pneumoperitoneum** is CO_2 (carbon dioxide) insufflation at 10–15 mmHg. CO_2 is used because it is non-combustible and highly blood-soluble, so embolism is less dangerous. Its effects have **two drivers**: **pressure** (compression of cava, renal veins and diaphragm) and **absorbed CO_2** (hypercarbia, acidosis, sympathetic drive).

Effect (supine)	10 mmHg	20 mmHg	40 mmHg
Heart rate	↑	↑	↓
MAP (mean arterial pressure), SVR (systemic vascular resistance)	↑	↑	↑
Cardiac output	↔/↑	↔/↓	↓
GFR (glomerular filtration rate), urine output	↓	↓↓	↓↓
Arterial PCO_2	↑	↑	↑

- **Cardiovascular:** afterload rises (aortic compression, catecholamines, vasopressin). Preload depends on volume: in hypovolaemia IVC (inferior vena cava) compression lowers venous return. **CVP (central venous pressure) reads falsely high.** Hypercarbia gives tachycardia and ectopics; peritoneal stretch gives **vagal bradycardia**.
- **Respiratory:** splinted diaphragm lowers FRC (functional residual capacity) and compliance, raises airway pressure; atelectasis. Barotrauma (pneumothorax) at high pressure.
- **Acid–base:** absorbed CO_2 causes respiratory acidosis; raise ventilation. In COPD (chronic obstructive pulmonary disease) check arterial gases; **hypercarbia peaks after extubation.**
- **Renal: oliguria** from **renal vein and parenchymal compression**; reversible on desufflation.
- **Others:** splanchnic flow falls; ICP (intracranial pressure) and intraocular pressure rise; smaller stress and immune response than open surgery; hypothermia; shoulder-tip pain.
- **Position:** head-down raises MAP and output but worsens lungs; head-up lowers output. Retroperitoneal insufflation causes smaller haemodynamic swings.
- **Safe practice:** preoperative cardiopulmonary assessment; optimise volume; **work at 10–12 mmHg**, 20 mmHg only briefly; monitor end-tidal CO_2, airway pressure and urine output.
- **Also know:** helium or argon for lung disease (poorly soluble, so embolism risk); low-pressure surgery with valveless insufflators.

Part B: Laparoscopic donor nephrectomy

Introduction

- First described by **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first** at every stage.

Donor evaluation

- Questionnaire → physical and psychosocial review, blood group, HLA (human leukocyte antigen) typing → **GFR by at least two methods, one measured** → metabolic, serology, cancer screening → **CT (computed tomography) angiography** for vessels → cardiac review if needed.

- Exclusions: kidney disease, malignancy, active infection, cardiovascular disease, diabetes, significant obesity; bilateral stones generally exclude.

Side selection

- **Left preferred:** longer renal vein, simpler dissection. **Right** if the left has multiple arteries or complex veins, or the right kidney is smaller: **the donor keeps the better kidney.**
- **Anatomy matters:** arterial variation in 25–40%; early (prehilar) branching and accessory arteries must be seen on CT, as they may contraindicate the laparoscopic route.

Techniques

Technique	Features
Pure transperitoneal	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds)	Hand port doubles as extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	No bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed instruments; single incision, less pain

Operative steps (left)

1. Reflect colon; divide **gonadal, adrenal and lumbar veins**.
2. Preserve periureteric tissue and the **gonadal–ureter triangle** (ureteric blood supply).
3. Keep pressure ≤10 mmHg; hydration and mannitol or furosemide to keep urine >100 mL/hour (counters pneumoperitoneum oliguria).
4. Divide ureter, then **artery first, then vein**. Vein stapled lateral to the adrenal vein; Endo-TA (non-cutting) stapler gives extra vein length.
5. Extract quickly; cold flush. Warm ischaemia a few minutes.

- **Hem-o-lok:** manufacturer contraindicates it on the donor renal artery after stump bleeds.
- **Open donor nephrectomy** is kept for complex anatomy or previous surgery, more often on the right.

Advantages versus open

- Less pain, shorter stay, faster return to work, better cosmesis; **more people volunteer to donate**; equal graft function.

Disadvantages and complications

- Longer operation, learning curve, shorter vessels (right vein), slightly longer warm ischaemia.
- Complications about **16%** (mostly gastrointestinal); Clavien ≥IV 2.5%; **mortality 3 per 10,000**; conversion 0.26–3.34%, mainly for vascular injury.

Long-term donor counselling

- **Also know:** kidney paired donation for ABO-incompatible or crossmatch-positive pairs; living-donor grafts survive longer than deceased-donor grafts.
- Function 60–80% of pre-donation; lifetime ESRD (end-stage renal disease) **90 per 10,000** versus 14 in matched non-donors (Muzaale, 2014); more gestational hypertension.

Diagram to draw: left renal hilum with gonadal, adrenal and lumbar veins, stapled artery and vein, and the gonadal–ureter triangle.

Key points to remember:

- Pneumoperitoneum: pressure plus CO_2; afterload rises, oliguria from renal vein compression.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Q3a. Nocturnal enuresis

Type: SAQ (10 marks) (also answered in Practice Set 1 Paper 3, Q3e)

Definition (ICCS [International Children's Continence Society])

- Intermittent incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

Epidemiology

- About **5–10% at 7 years**; falls with age; about 2% of adults; familial.

Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (poor night-time ADH [antidiuretic hormone] rise).
2. **Small nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep.

- **Why sleep apnoea matters:** obstructed breathing makes arousal difficult and releases atrial natriuretic peptide, which raises urine output.
- Comorbidities: constipation, sleep-disordered breathing, ADHD (attention deficit hyperactivity disorder).

Evaluation

- History, examination (spine, genitalia, abdomen), urinalysis.
- **Bladder diary:** voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL; night output by weighing nappies.
- Uroflowmetry, PVR (post-void residual), ultrasound only for NMNE or before anticholinergics.

Management

- **Treat daytime LUTS and constipation first.** Education; waking, fluid restriction and punishment do not work.

	Alarm	Desmopressin
Target	Arousal (conditioning)	Night urine output
Use	Nightly; until 14 dry nights	0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed
Response	50–70%; durable	Rule of thirds; relapses off drug
Caution	Motivated family	Limit evening fluid (hyponatraemia)

- **Failure:** switch or combine. **Second line:** add oxybutynin after checking PVR and bowels. **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists, adenotonsillectomy for sleep apnoea.
- **Adults:** new enuresis may signal high-pressure chronic retention: catheterise and treat obstruction.

Diagram to draw: triad: nocturnal polyuria, small capacity, arousal failure.

> **Key points to remember:**
> - ≥5 years; secondary after 6 months dry.
> - Alarm (durable) and desmopressin (rule of thirds) are first line.
> - Treat daytime symptoms and constipation first.

Q3b. BXO

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 1, Q3c)

BXO (balanitis xerotica obliterans) is taken as genital lichen sclerosus (LS) in the male.

Definition

- Chronic, progressive, **scarring inflammatory disease of genital skin**; possibly premalignant. BXO is its late stage in men.

Epidemiology and aetiology

- 6–10 times commoner in females; in men, older men and boys with phimosis. 15–20% have extragenital disease.
- Cause unknown. Autoimmune links (thyroid disease, vitiligo), familial clustering; **chronic occlusion by urine** under the foreskin (dribbling, obesity) definitely increases risk. Infection and trauma are proposed but unproven.

Clinical features

- White, atrophic sclerotic plaques with loss of architecture, purpura, fissures; itch, painful erection, poor stream, spraying.
- Sequence: prepuce → **phimosis** → glans → **meatal stenosis** → fossa navicularis and penile urethral stricture.
- **Spares the bulbar and posterior urethra.**
- **Stricture mechanism:** voiding against meatal stenosis forces urine into the glands of Littre → inflammation → spongiofibrosis.

Evaluation

- **Biopsy** if doubtful: hyperkeratosis, epidermal atrophy, basal vacuolation, homogenised dermal collagen, band-like lymphocytes.
- Uroflowmetry; **RUG (retrograde urethrogram)** before meatal surgery, because a proximal stricture may coexist.
- **Differential:** lichen planus, Zoon balanitis, penile intraepithelial neoplasia.

Management

Situation	Treatment
Skin disease	**Clobetasol propionate 0.05%** for about 3 months, then twice-weekly maintenance; calcineurin inhibitors second line
Phimosis	**Circumcision**, usually curative
Meatal stenosis	Meatotomy; BMG (buccal mucosal graft) meatoplasty
Anterior stricture	**Oral mucosa, never genital skin** (LS recurs in skin)
Panurethral or unfit	**Perineal urethrostomy** (easy because the proximal urethra is spared); self-dilatation

- Recurrence is common after medical treatment, and restricture rates exceed other aetiologies; treatment follows the patient's goals.

Malignancy

- Penile SCC (squamous cell carcinoma) in **2.3–9%**, HPV (human papillomavirus)-independent type. **Lifelong follow-up**; biopsy any ulcer or nodule.

Diagram to draw: penis showing prepuce, glans, meatus and fossa involved, proximal urethra spared.

Key points to remember:

- Meatus is the sentinel site; proximal urethra spared.
- Clobetasol first; oral mucosa, never skin.
- SCC in 2.3–9%: lifelong follow-up.

Q3c. Autotransplantation

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 3, Q3c and 2025-Winter Paper 3, Q3c)

Taken as renal autotransplantation.

Definition and principle

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** if needed. First done in 1963 for proximal ureteric injury.
- Bringing the kidney to the bladder replaces the ureter without bowel.

Indications

Group	Examples
Ureteric loss (commonest)	Long iatrogenic or traumatic loss; failed repairs (**delayed setting, never acute**)
RPF (retroperitoneal fibrosis)	Ureterolysis impossible
Tumour	UTUC (upper tract urothelial carcinoma) in a solitary kidney; bench partial nephrectomy
Vascular	Complex renal artery aneurysm or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker
Pain	Selected loin pain–haematuria syndrome

- **Preferred over ileal ureter** when the other kidney is poor or bowel unsuitable (creatinine >2 mg/dL, bowel disease, radiation).
- **Contraindications:** poor renal function, severe iliac atherosclerosis, unfit patient.

Procedure

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CT (computed tomography) angiography, renogram
→ harvest as for live donation: maximal vessel length,
  preserve ureteric blood supply
→ cold perfusion ± bench repair
→ Gibson incision, extraperitoneal iliac fossa
→ vein to external iliac vein (end-to-side);
  artery to external iliac (end-to-side) or internal iliac (end-to-end)
→ pyelovesicostomy (± Boari flap), ureteroneocystostomy,
  or native ureter left in continuity
→ stent, catheter, drain
```

- Laparoscopic harvest speeds recovery; robotic autotransplantation emerging.

Complications and outcome

- Arterial or venous thrombosis, bleeding, urine leak, anastomotic stricture, lymphocele, infection.
- **Follow-up:** Doppler ultrasound early; renogram and blood pressure later.
- **Renal unit loss 5–10%**; major (Clavien–Dindo ≥III) complications about 15–20%; **success about 90%**.

Also know: alternatives: Boari flap, TUU (transureteroureterostomy), ileal ureter, Yang–Monti.

Diagram to draw: kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

Key points to remember:

- For long ureteric loss when the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for vascular or tumour cases.
- Unit loss 5–10%; success about 90%.

Q3d. Congenital Penile curvature..

Type: SAQ (10 marks)

Definition and principle

- Curvature of the erect penis present from birth, with an **orthotopic meatus**.
- Rigidity needs **symmetric compliance** of the tunica albuginea; curvature means one aspect is less compliant.

Embryology

- DHT (dihydrotestosterone) drives fusion of the urethral folds and maturation of spongiosum, Buck's and dartos fascia. The urethra may form normally while the mesenchyme stays deficient → dysgenetic, inelastic ventral tissue.
- Normal fetal ventral curvature resolves by 20 weeks.

Devine and Horton classification (1973)

Type	Abnormal tissue
I	Corpus spongiosum deficient (chordee without hypospadias)
II	Dysgenetic Buck's and dartos fascia; normal spongiosum
III	Dartos only
IV	Normal layers; **asymmetric corporal compliance** (true congenital curvature)
V	Congenitally short urethra (very rare; doubted)

Two clinical groups

Feature	Chordee without hypospadias (I–III)	Congenital curvature (IV)
Direction	Ventral ± torsion	Ventral, lateral (often left), rarely dorsal
Erection size	In proportion to flaccid size	**Noticeably large**
Presentation	Childhood	18–30 years, noticed at puberty

- **Versus acquired** (Peyronie disease, fracture): congenital is painless, no plaque, no shortening.

Evaluation

- Home photographs of erection; **artificial erection** (Gittes–McLaughlin saline test or prostaglandin) in theatre.
- Grade: mild <30°, moderate 30–45°, severe ≥45°.

Management

- Surgery if curvature hinders intercourse or distresses.

1. Deglove; excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum.
2. Residual curve: **midline ventral septotomy** under artificial erection.
3. Still curved: **dorsal plication** contralateral to the curve: Nesbit (excise ellipses, close transversely), Yachia (Heineke–Mikulicz), 16-dot or TAP (tunica albuginea plication). **12 o'clock is nerve-free**, so midline plication is safe.
4. Severe: ventral **corporotomy with graft** (dermis, tunica vaginalis).

- **Rule:** if the meatus reaches the glans tip, do not divide the urethra; the corpora usually limit straightening.
- **Results:** single operation in almost all; recurrence 2–15%, mostly after plication.

Diagram to draw: penile cross-section: nerves at 11 and 1 o'clock, plication at 12.

Key points to remember:

- Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
- Congenital curvature: young man, large erection, orthotopic meatus.
- Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

Q3e. Maturation Arrest.

Type: SAQ (10 marks)

Taken as spermatogenic maturation arrest on testicular histology.

Definition

- A histological pattern of **spermatogenic failure** in which germ cells develop normally up to one stage and **the sequence halts** at a premeiotic, meiotic or postmeiotic step. No mature sperm, or very few, are made.
- One of three patterns in NOA (non-obstructive azoospermia):

Pattern	Histology
Hypospermatogenesis	All stages present, fewer cells
Maturation arrest	Halts at one stage
SCO (Sertoli cell–only)	No germ cells

Classification

- **Early:** arrest at spermatogonia or **primary spermatocyte** (meiotic failure).
- **Late:** arrest at the **spermatid** stage.
- **Complete** (all tubules) versus **incomplete** (foci progress to sperm): incomplete gives better retrieval.

Causes

- **Genetic:** complete **AZFb (azoospermia factor b) deletion** → arrest; meiotic recombination defects in about half of men with maturation arrest; karyotype anomalies.
- **Acquired:** heat (varicocele, cryptorchidism), chemotherapy, radiation, drugs, orchitis; idiopathic often.
- Intratesticular testosterone is essential for postmeiotic maturation.

Clinical features and diagnosis

- Azoospermia or severe oligospermia, normal semen volume, palpable vasa.
- **Testis size and FSH (follicle-stimulating hormone) are often normal**, so it can mimic obstruction (FSH ≤7.6 IU/L, long axis >4.6 cm).
- **Diagnosis needs testicular biopsy** (Bouin's fixative, never formalin), usually done as sperm extraction.
- **Karyotype and Y microdeletion** testing in all azoospermic men (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine]).

Management

- Optimise: treat varicocele, stop exogenous testosterone; off-label hormonal stimulation (clomiphene, hCG [human chorionic gonadotropin]) unproven.
- **Micro-TESE (microdissection testicular sperm extraction)** with ICSI (intracytoplasmic sperm injection): retrieval **about 40%** (versus 30% in SCO).
- **Complete AZFb deletion: retrieval highly unlikely; do not operate.**
- No sperm: donor sperm or adoption. Round spermatid injection is experimental.
- Genetic counselling: sons inherit Y deletions.

Diagram to draw: seminiferous tubule showing germ cells up to spermatocytes with no spermatids or sperm.

> **Key points to remember:**
>
> - Arrest at one stage; early (spermatocyte) or late (spermatid).
> - FSH and testis size may be normal: biopsy diagnoses.
> - Micro-TESE retrieval about 40%; none with complete AZFb deletion.

Q3f. Detrusor sphincter dyssynergia

Type: SAQ (10 marks) (also answered in 2016-Summer Paper 1, Q3a)

Definition

- **DSD (detrusor–external sphincter dyssynergia):** involuntary contraction of the striated sphincter during detrusor contraction, causing functional obstruction, high pressure and poor emptying.
- **Lesion between the pons and the sacral cord:** suprasacral SCI (spinal cord injury), MS (multiple sclerosis), transverse myelitis, myelomeningocele. **Never with brain lesions** (stroke, Parkinson's).
- Lesions at or above T6 may add **smooth sphincter (bladder-neck) dyssynergia** and autonomic dysreflexia.

Blaivas classification

Type	EMG (electromyography) pattern	Lesion
1	Activity rises with detrusor pressure, relaxes suddenly at peak	Incomplete
2	Sporadic contractions throughout voiding	Complete
3	Crescendo–decrescendo; obstruction throughout	Complete

- Also intermittent versus continuous. Type is unrelated to level.

Diagnosis

- **Pressure–flow study with EMG:** EMG rises with the detrusor contraction; high pressure, interrupted flow. Needle EMG is more reliable.
- **VUDS (videourodynamics):** narrow membranous urethra, dilated posterior urethra; VUR (vesicoureteral reflux).
- **Exclude:** guarding reflex (normal); **pseudodyssynergia** (straining or suppressing a contraction); dysfunctional voiding (learned, no neurological lesion).

Significance

- Worse when complete, continuous and **in men**: hydronephrosis, VUR, stones, urosepsis. Other risk factors: high filling pressure, indwelling catheter.
- **Management logic:** DSD is both a storage and an emptying failure. Lower bladder pressure, then empty it.
- DLPP (detrusor leak point pressure) ≥40 cmH_2O endangers the kidneys.

Management

- **First line:** CIC (clean intermittent catheterisation) plus antimuscarinic or β3-agonist; then intradetrusor botulinum toxin.
- **CIC impossible (men):** sphincterotomy or sphincteric botulinum toxin with condom drainage; DLPP ≥40 afterwards means failure.
- Others: suprapubic catheter, Brindley stimulator with deafferentation, diversion. Urethral stents abandoned. Credé and Valsalva voiding contraindicated.
- **Also know:** α-blockers for bladder-neck dyssynergia; lifelong upper-tract surveillance.

Diagram to draw: urodynamic trace with rising detrusor pressure, EMG bursts and interrupted flow.

> **Key points to remember:**
>
> - Lesion between pons and sacral cord; never brain lesions.
> - Blaivas 1 incomplete; 2 and 3 complete.
> - Worst in men; DLPP ≥40 cmH_2O hostile.

Rapid Revision Sheet – Practice Set 10 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Management of chyluria (LAQ)

- Pyelolymphatic fistula from *W. bancrofti* lymphatic obstruction; often intermittent.
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- Diet with MCT plus DEC → silver nitrate sclerotherapy → hilar lymphatic disconnection.
- Retroperitoneoscopic disconnection beats open.

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- Pneumoperitoneum: pressure plus CO_2; afterload rises, oliguria from renal vein compression.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Q3a. Nocturnal enuresis (SAQ)

- ≥5 years; secondary after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

Q3b. BXO (SAQ)

- Meatus is the sentinel site; proximal urethra spared.
- Clobetasol first; oral mucosa, never skin.
- SCC in 2.3–9%: lifelong follow-up.

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- For long ureteric loss when the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for vascular or tumour cases.
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- Arrest at one stage; early (spermatocyte) or late (spermatid).
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- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 cmH_2O hostile.